

IVÁN ALEJOS GÓMEZ

PIPELINE SOFTWARE
DEVELOPER | PYTHON & 3D
TOOLING

CONTACT

- +34 671 18 48 64
- ivanalejosgomez99@gmail.com
- Valencia, Spain (Torrent)
- English (Professional / Fluent)

PIPELINE & TOOLS

Python (OOP)

Qt / PySide (UI)

C++ Integration

DCC Tooling

Houdini (VEX / Python)

Autodesk Maya

USD Concepts

REST APIs / Postman

SYSTEMS & DATA

Git / GitLab

Linux (Ubuntu)

Windows / macOS

PostgreSQL / SQL

Bash / Shell Scripting

Agile / Scrum

LANGUAGES

Spanish **Native**
English **Professional / Fluent**

PROFESSIONAL PROFILE

Multimedia Engineer with a Master's Degree in Computer Graphics, Video Games & VR, specializing in **animation & VFX pipeline software development, Python tooling, DCC integration, and workflow automation**. Proven track record building custom desktop tools, APIs, and data integrations bridging technical engineering with creative production teams. Proficient in **Python, C++, Qt/PySide, Git, and REST APIs**, with hands-on expertise in 3D graphic pipelines and procedural systems (**Houdini, Maya, Unity, OpenGL**). Passionate about optimizing production environments, cross-platform stability (Linux/Windows), and building intuitive tools that empower artists to create efficiently.

WORK EXPERIENCE

● Pipeline & Systems Software Developer / Technical Specialist

Jul. 2024 - Jul. 2026

SIMODAR TRAINING SOLUTIONS SL | Paterna, Valencia

- **Pipeline & Tooling Development (Python / C++):** Designed, developed, and maintained custom software tools, desktop interfaces (Qt/PySide), and modular libraries to automate data flow and hardware/software integration.
- **DCC & Real-Time Engine Integration:** Built automated asset pipelines, communication protocols (REST APIs / WebSockets), and data converters between 3D DCC environments, real-time simulation engines, and sensor hardware.
- **Cross-Platform Architecture (Linux & Windows):** Maintained and optimized pipeline tools ensuring robust, reliable execution across multiplatform operating system environments.
- **Technical Problem Solving & Artist Enablement:** Diagnosed and resolved complex production pipeline bottlenecks in real-time, working closely with technical and creative teams to translate operational needs into clean software solutions.
- **Version Control & Agile Collaboration:** Managed code repositories, modular releases, and branching workflows using Git/GitLab in fast-paced collaborative sprints.

● Software Developer, Automation & Systems Engineer

Abr. 2021 - Mar. 2024

Diseño e Innovación Técnica SL | Paterna, Valencia

- **Workflow Automation & Internal Tooling:** Engineered automated Python and SQL scripts for data processing, batch asset management, telemetry recording, and server synchronization.
- **Database & Production Tracking Integration:** Designed, migrated, and maintained relational database schemas (PostgreSQL, SQL Server) for production tracking, user permissions, and digital asset metadata.
- **GUI & Service Development:** Developed graphical user interfaces and background services connecting internal APIs, video streams, and enterprise systems (Odoo / SCADA).
- **Infrastructure & Server Maintenance:** Administered Linux and Windows server environments, managing network storage and deployment reliability.

● Multimedia & Web Integrator

Feb. 2021 - May. 2021

Cubicup Design Home SL | Valencia

- **Asset Integration:** Digital asset management, UI design, and technical integration.

EDUCATION & ACADEMIC BACKGROUND

Master's Degree in Computer Graphics, Video Games & VR

Universidad Rey Juan Carlos | Madrid, Spain (Sept. 2023 - Jun. 2025)

Master's Thesis (TFM): Advanced Fluid Simulation in **Houdini** using Particle-In-Cell methods, procedural VEX programming, numerical dynamics, and rendering optimization. Advanced studies in computer graphics architectures, procedural generation, rendering pipelines, and C++ optimization.

Bachelor's Degree in Multimedia Engineering (Grado en Ingeniería Multimedia)

Universitat de València | Valencia, Spain (Sept. 2017 - Jun. 2022)

Comprehensive foundation in Software Engineering, Computer Graphics, Operating Systems (Linux), Relational Databases, and Human-Computer Interaction (UI/UX).